

Author Response to Reviewer tcvY

All changes marked in blue; line refs refer to revised `main.tex`.

R2.1: Insufficient 3D GNN baselines (only SchNet). **Response:** Added PaiNN as second baseline: Section 5.4 [L427--453], Table 5 [L432--449], Appendix L [L1066--1087]. PaiNN (equivariant message passing, single conformer) achieves 28%/42%/30% gains on ESOL/FreeSolv/Lipo—convergence with SchNet’s 21–42% across two architecturally diverse GNNs confirms the hierarchy is robust to architecture choice. Justification for omitting DimeNet++/Equiformer at [L453]. Re: ensemble pooling—our DKO mean ensemble [L351] already does this and underperforms FP+XGBoost. Abstract [L82], Contribution 3 [L157] updated.

R2.2: FreeSolv directionally inconsistent (−13.5% to +18.0%). **Response:** Clarified at [L566]: the +18.0% is a 3-seed random-split auxiliary on $N=642$; the definitive result is the 10-seed scaffold-split t -test: −13.5%, $p < 3 \times 10^{-5}$ (Table 4, [L518]). Learning curves [L569--571] show benefit emerges above ~ 440 molecules.

R2.3: Zero-padding/truncation to fixed D may be the bottleneck. **Response:** Feature dimension ablation [L484--508] (Table 6, [L489--504]): RMSE monotonically degrades $D=256 \rightarrow 2048$. If truncation were the bottleneck, larger D should help—it hurts. The limitation is the curse of dimensionality, not information loss from truncation. BDE’s variable dims (187–1,854) acknowledged; adaptive-length encoding noted as future work [L508]. This does not change the core finding that pre-computed statistics discard relational structure GNNs preserve.

R2.4: Excluding pre-trained models limits taxonomy. **Response:** Agreed. Taxonomy explicitly scoped to non-pre-trained setting: Related Work [L173], Limitations [L620]. Pre-trained 3D models (Uni-Mol) may shift taxonomy boundaries through learned representations trained on millions of conformers. DKO statistics are complementary—they could augment pre-trained embeddings—and testing this combination is noted as future work.

R2.5: Restriction to regression. **Response:** Added BACE/BBBP classification: Section 5.5 [L461--481], Table 7 [L466--479]. Conformer features hurt BACE (−4.4% AUROC), no effect on BBBP (+0.1%)—taxonomy confirmed for classification. Neural DKO variants evaluated on regression only (classification requires architectural adaptation); hybrid FP+conformer evaluated on both. Scope [L151], abstract [L80], Appendix K [L1041--1064].